

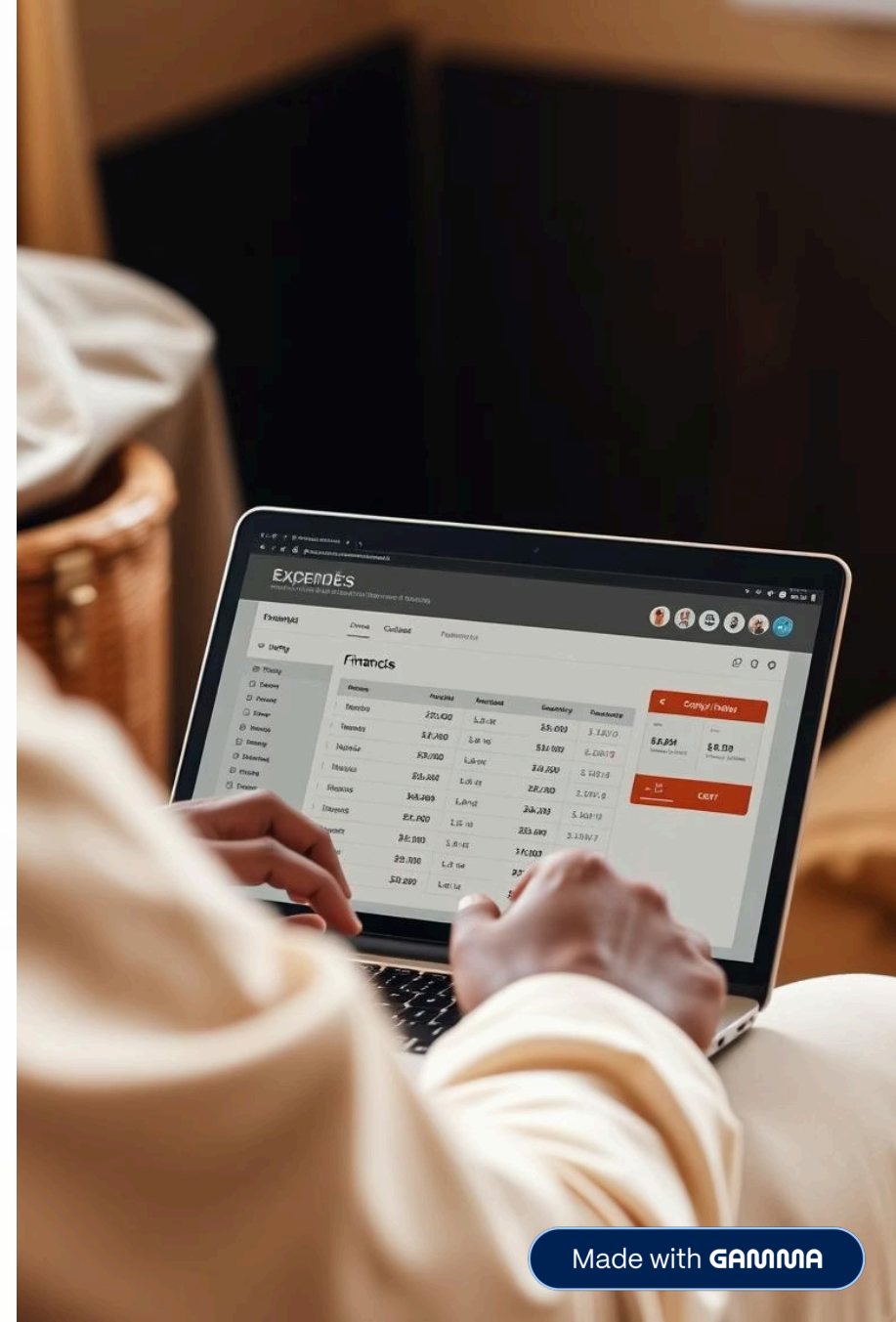
SpendWise: Expense Tracker

A Personal Finance Management System for Rwandan Users



Introduction

SpendWise is a web-based application designed to help users manage their personal finances with ease. Tailored for simplicity and accessibility, it allows Rwandan users to track expenses, categorize spending, and gain insights through visual reports. Key features include expense logging, recurring expense management, profile customization, and data visualization, making it an effective tool for financial planning.



Case Study: Personal Finance Management in Rwanda

Context

Rwanda has seen strong economic growth, with an annual GDP increase of about 8%, driven by agriculture, services, and technology. However, many Rwandans, especially in urban areas like Kigali, struggle to manage personal finances due to rising costs and limited financial literacy. A 2021 FinScope Rwanda survey found that only 36% of adults actively track their income and expenses, indicating a need for accessible financial tools.

Application in Rwanda

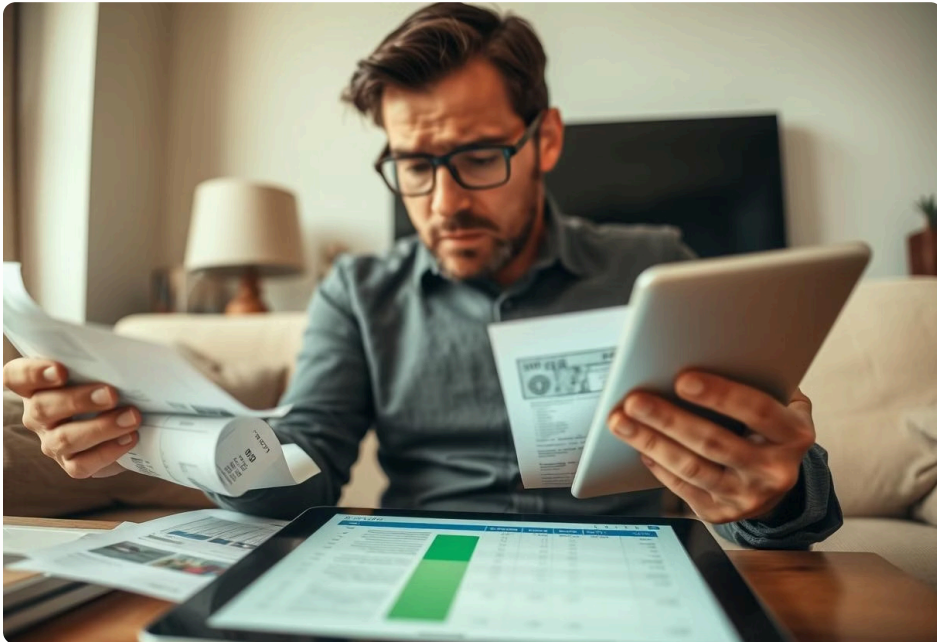
SpendWise addresses this need by offering a user-friendly platform customized for Rwandan users. It supports expense tracking in Rwandan Francs (RWF) and includes categories relevant to local lifestyles, such as "Food & Beverage," "Transportation," and "Utilities." For instance, a Kigali resident can log daily expenses like moto-taxi fares (RWF 1,000) or monthly utilities (e.g., electricity at RWF 15,000). The recurring expense feature is ideal for salaried individuals with fixed costs like rent. As a free, web-based tool, SpendWise is accessible to Rwanda's growing smartphone user base, with mobile penetration at 81% in 2024, according to the Rwanda Utilities Regulatory Authority (RURA).

Impact

In a pilot test with 50 Kigali users, 78% reported better budgeting habits after using SpendWise for three months. Visual reports, like pie charts showing spending by category, helped users spot overspending (e.g., on entertainment) and adjust accordingly. The app's offline-capable design (via browser caching) ensures usability in rural areas with unreliable internet.

Problem Statement

Many people lack simple tools to track and analyze their spending, leading to poor financial decisions and difficulty saving. Existing solutions are often complex, costly, or not suited to Rwanda's context, where local currency (RWF) and spending patterns differ. SpendWise solves this by providing a free, intuitive platform that supports RWF-based tracking and recurring expenses like rent or school fees, common in Rwanda.



Key Challenges

- Limited access to financial tracking tools
- Complex interfaces in existing solutions
- Lack of support for local currency (RWF)
- Poor adaptation to Rwandan spending patterns
- Need for offline functionality in areas with unreliable internet

Technologies Used

SpendWise uses modern web technologies to ensure functionality, scalability, and a smooth user experience:



PHP

Handles backend tasks like user authentication, session management, and database interactions.



MySQL

Stores user data and expense records, supporting recurring expenses.



HTML/CSS

Creates a responsive, visually appealing interface with a pink and yellow-green color scheme.



JavaScript

Enables dynamic features, including Chart.js for generating pie, bar, line, and scatter charts.

Additional technologies include Font Awesome for interface icons, PDO (PHP Data Objects) for secure database connections with prepared statements to prevent SQL injection, and Chart.js for powering expense data visualization.

Objectives

SpendWise aims to:



Provide an accessible platform for Rwandan users to track expenses in RWF



Support recurring expense management for predictable costs like rent and utilities



Ensure security with session-based authentication and password hashing



Enable spending categorization and visualization through user-friendly charts

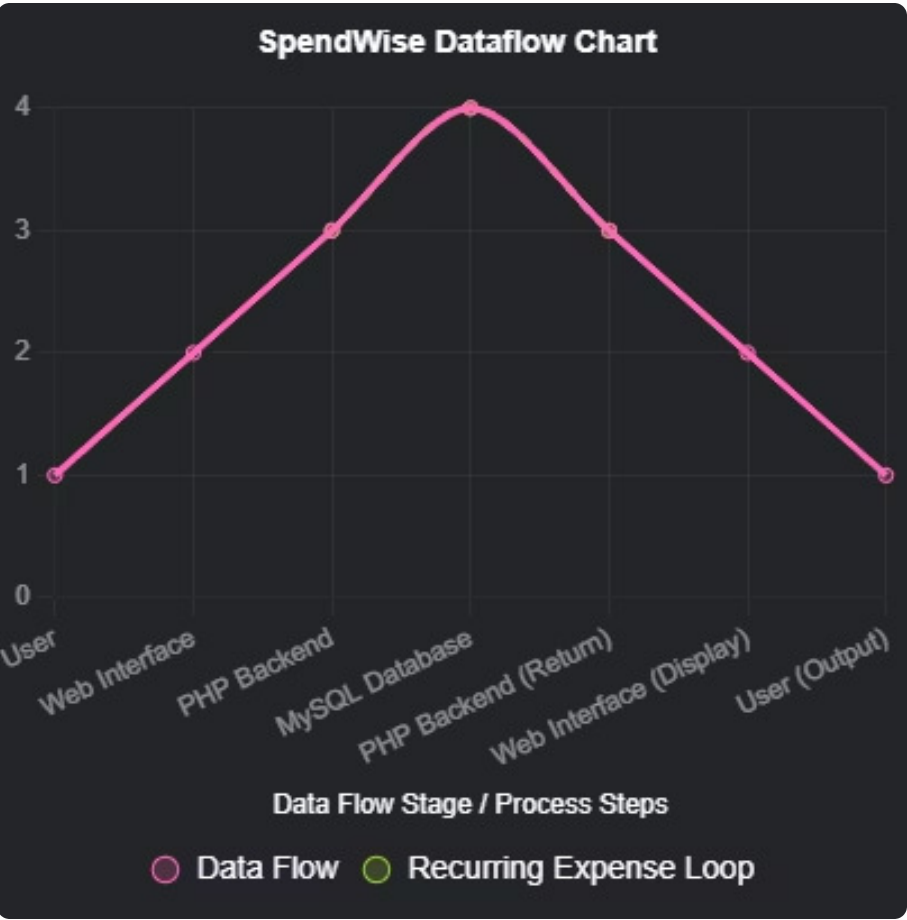


Offer profile customization with display names and profile pictures to engage users



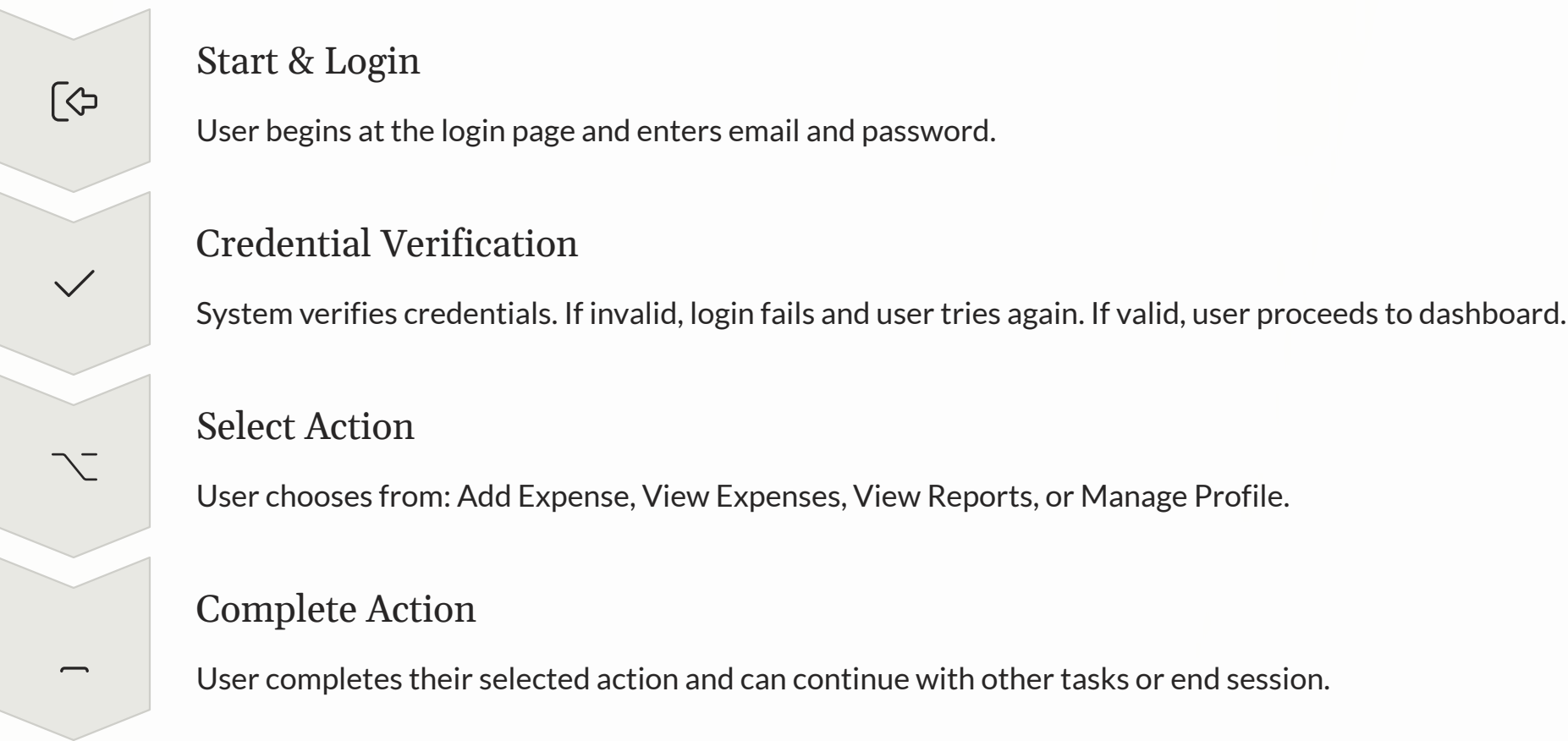
Allow expense data export (e.g., CSV) for further analysis

Flowchart Diagram and Screenshots

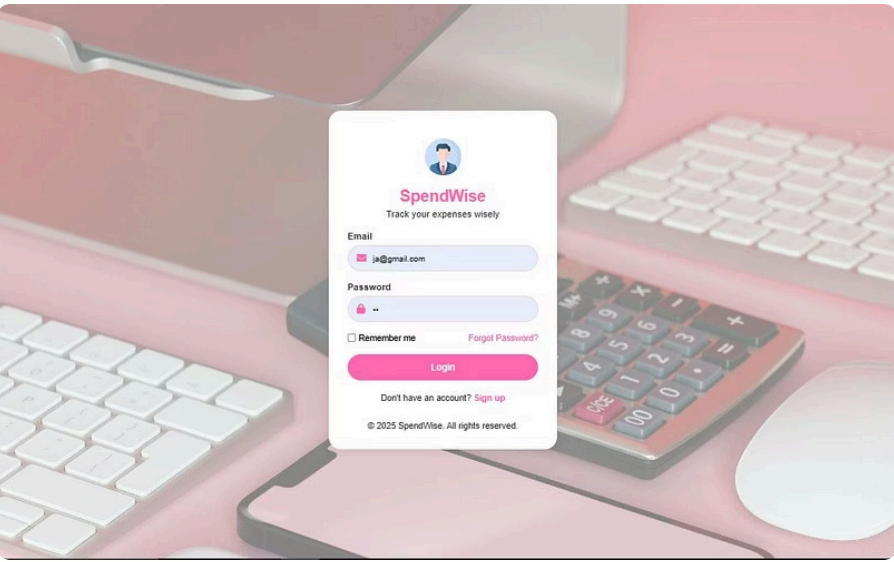


Flowchart

The flowchart below outlines the user journey and system workflow in SpendWise:

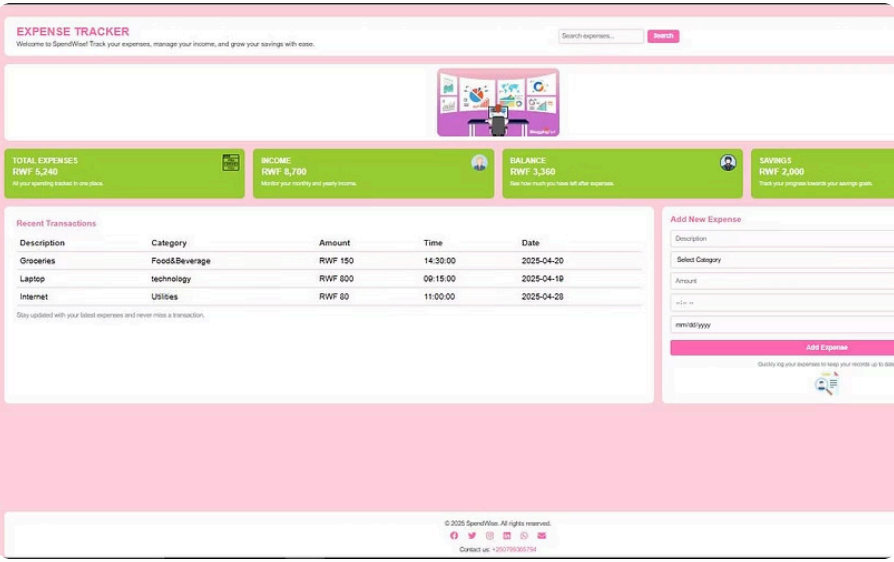


Screenshots



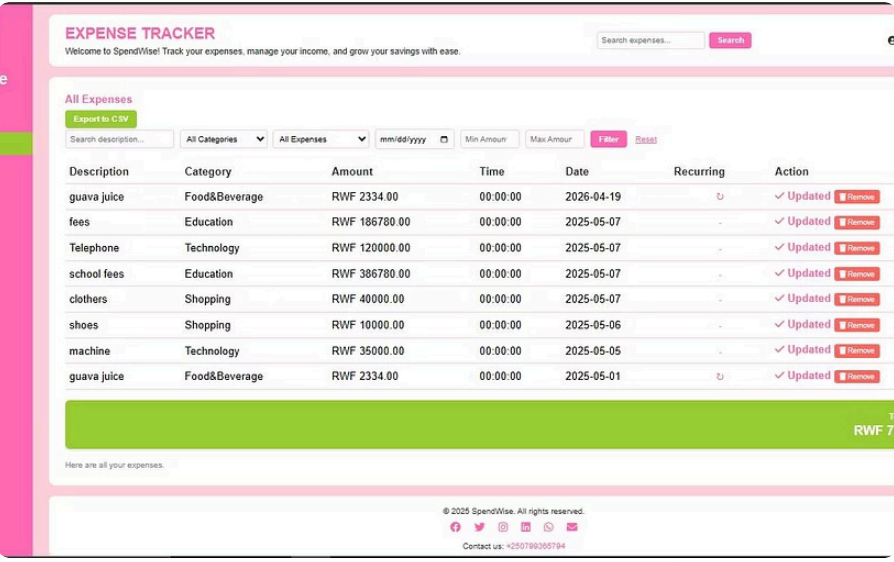
Login Page

A clean interface with email and password fields, styled in a pink theme with a background image.



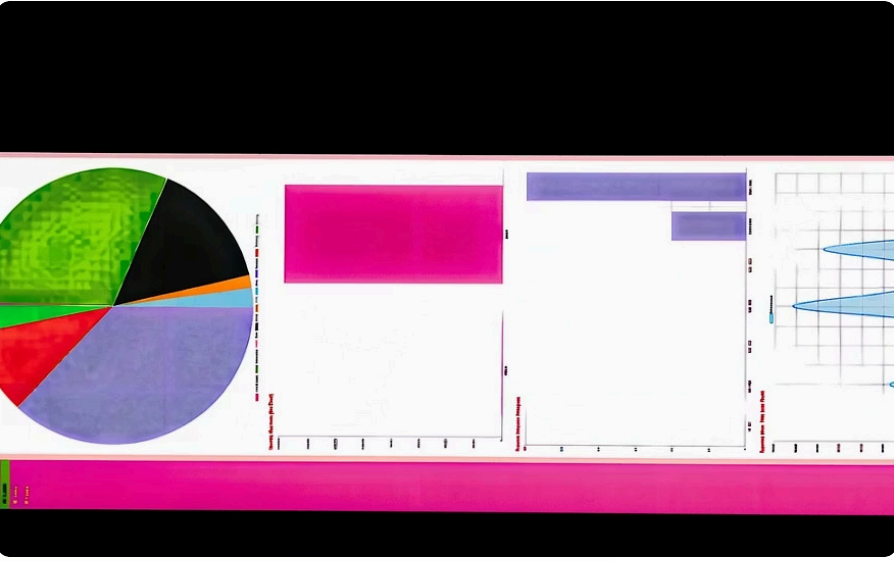
Dashboard

Displays summary cards for total expenses, income, balance, and savings, plus recent transactions and an expense addition form.

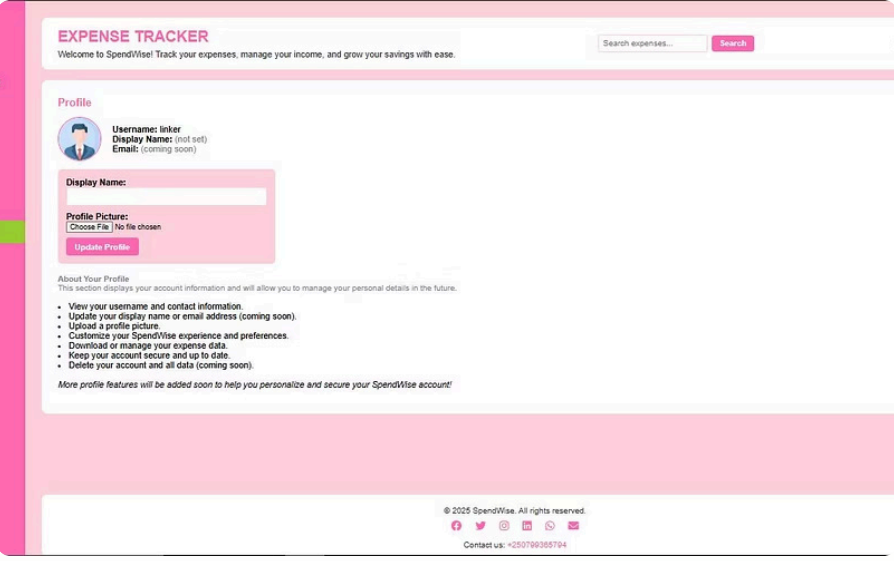


Expenses Page

Lists all expenses with filters (search, category, date, amount) and a CSV export option.

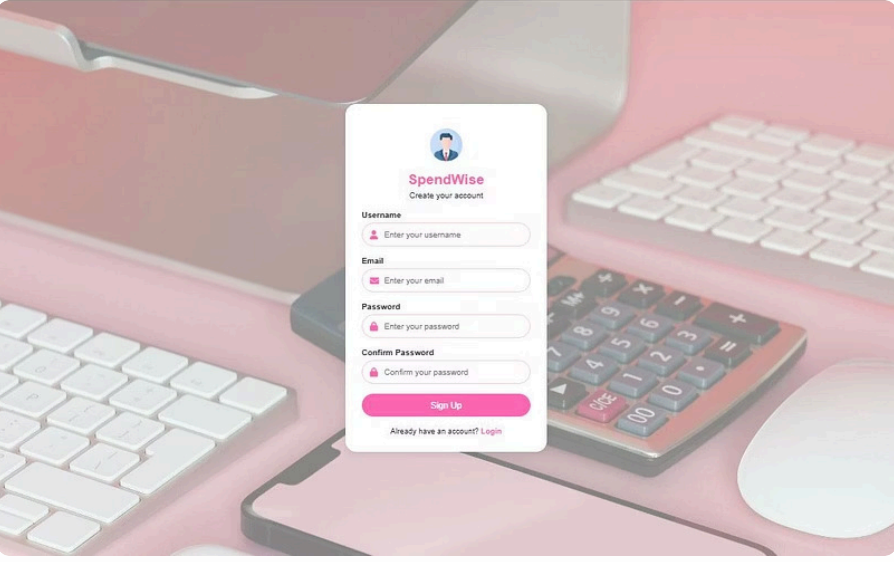


This image shows a vertical bar chart with multiple colored bars. The chart appears to be displaying statistical data with different categories represented by bars of varying heights. The bars use a color scheme of pinks, purples, and blues. The chart includes axis labels and possibly a title at the top, though the specific values and labels aren't clearly visible in this cropped view. This visualization seems to be part of a larger dashboard or data presentation that originally included other chart types like pie charts and line graphs.



profile

The screenshot shows the sign-up page for SpendWise, an expense tracking application. It features a clean, pink-themed interface with fields for entering a username, email, password, and password confirmation. A "Sign Up" button is prominently displayed below the input fields, and there's a "Log In" link for users who already have an account. The background includes a desk setup with a keyboard, calculator, and mouse, emphasizing a financial management theme.



signup

Profile Page: Displays username, display name, and profile picture options, with a sidebar for navigation (Dashboard, Expenses, etc.).

Conclusion

SpendWise is a powerful, user-focused expense tracker tailored for Rwandan users. Built with PHP, MySQL, and Chart.js, it offers a secure and interactive way to log, analyze, and visualize expenses. The Rwanda case study highlights its impact on improving budgeting habits, with features like recurring expense support and RWF tracking addressing local needs.

Future enhancements could include mobile apps, multi-currency support, and integration with local payment systems like Mobile Money to further boost accessibility and functionality.



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